

# Multiplayer SPSU Tub Racing Game

UC-142-177

## Abstract / Introduction

- Multiplayer Bathtub Racing revives a well-known Southern Polytechnic State University tradition through a digital multiplayer experience built in Unity. The project extends a prior single-player tub racing game by adding online multiplayer
- Historically, SPSU engineering students designed and raced bathtub cars as capstone projects. Because the physical event is no longer allowed, Our team extended that tradition as software that preserves campus nostalgia while making it easier for alumni and current students to engage with the idea.
- The goal was to add networking and steam integration to deliver a multiplayer build that feels true to the legacy of tub racing while remaining practical for future teams to extend.
- Out of scope areas were also completed including improved AI.

## Materials and Methods

### Materials:

- Unity 2023.2.20f1
- GitHub Desktop
- Steam / Steamworks
- GoNet networking
- Windows 10/11
- Minimum hardware: 3GB storage, 8GB RAM

### Methods:

- Reviewed the prior single-player codebase to identify strengths, weaknesses, and refactor needs.
- Planned the work around a waterfall process and milestone-based handoff expectations.
- Implemented Steamworks lobby and relay support for session creation, friends, and chat.
- Integrated GoNet to synchronize player movement and race state without an external database or metaserver.

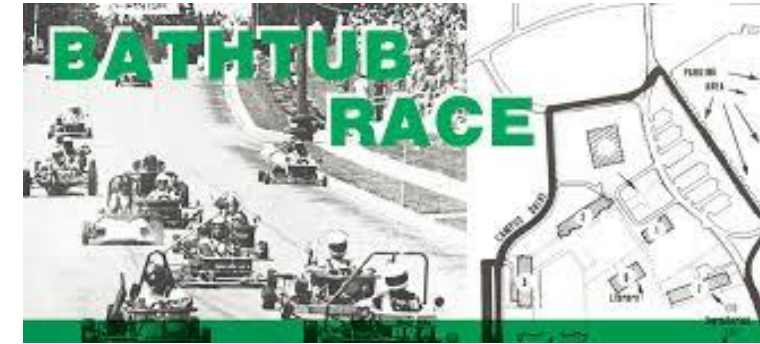
## Karting Highlights

### Why It Matters

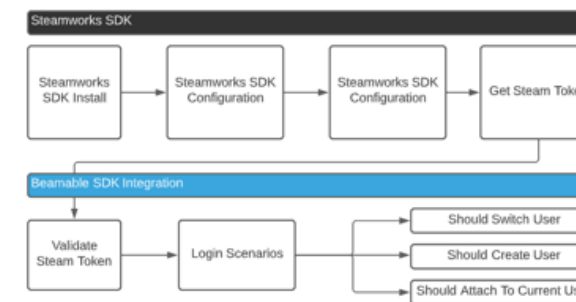
- SPSU engineers once designed and raced bathtub cars as student projects.
- The real event is no longer permitted, so the game becomes a digital revival of that tradition.
- Adding multiplayer turns last semester's prototype into a more social and alumni-facing experience.

### Key Implementation Choices

- Added a Steamworks lobby with friend support and in-game chat.
- Used Steam Relay for peer-to-peer play without port forwarding.
- Integrated GoNet for real-time race-state synchronization.

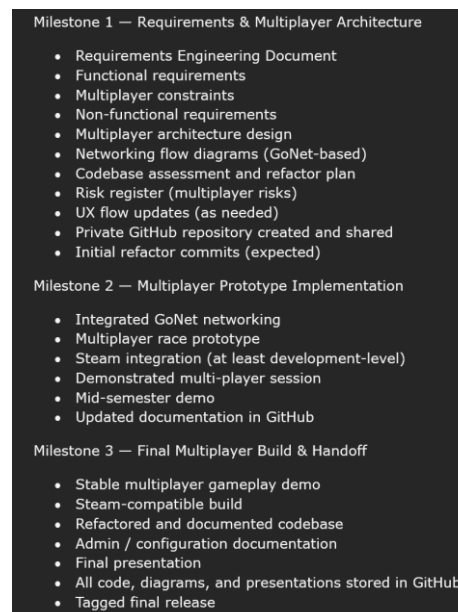


**Figure 1:**  
SPSU Tub  
racing History  
Artifact



**Figure 2:**  
Steamworks/SDK  
flow supporting  
authentication and  
multiplayer  
session setup.

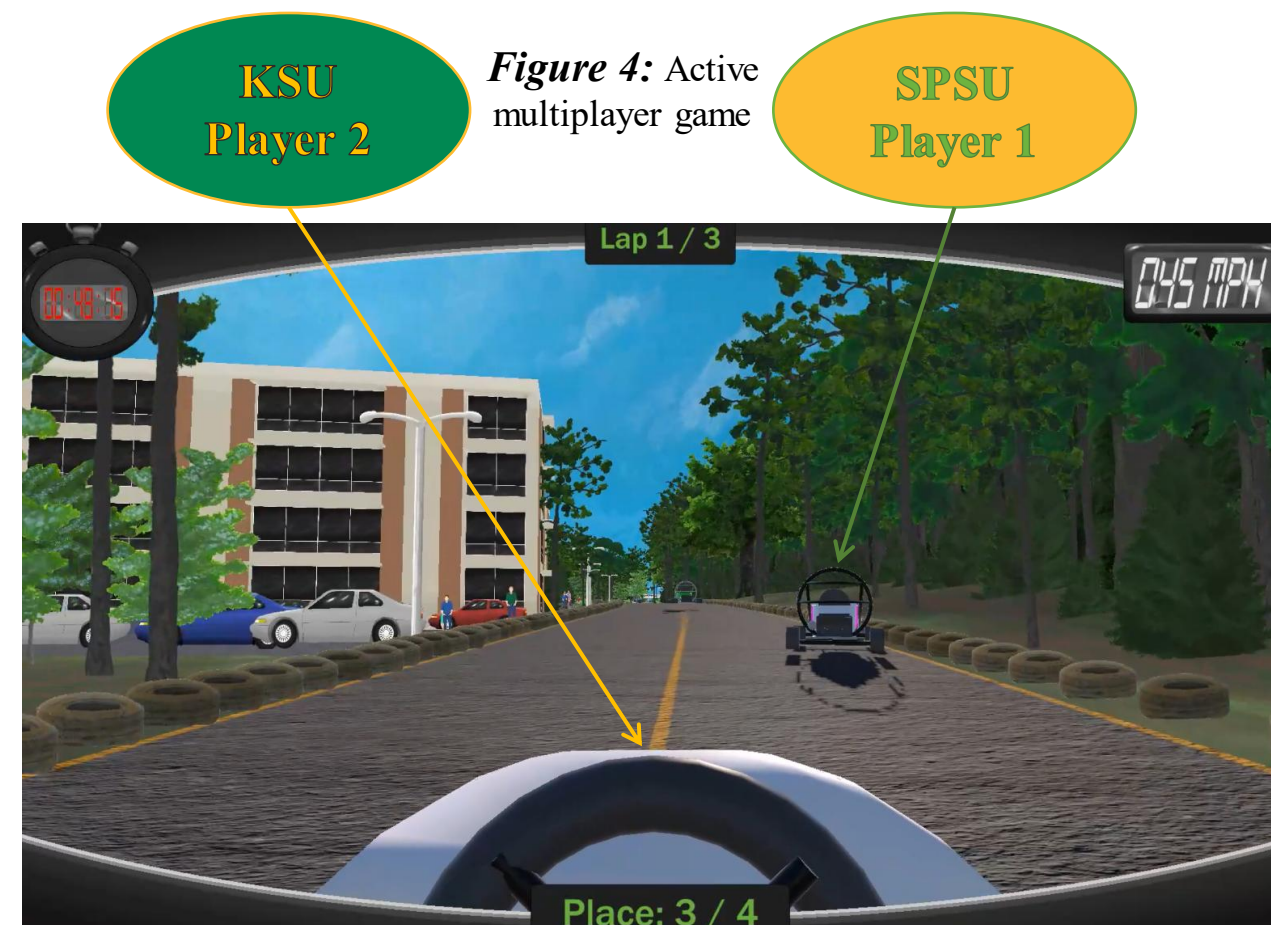
## Development Artifacts and Current State



**Figure 3:** Milestone-based original plan covering requirements, prototype implementation, and handoff.

## Project Website:

<https://multiplayertubracinggame.com/>



**Figure 4:** Active  
multiplayer game

**Figure 5:**  
Historical  
Southern Tech  
bathtub race poster



## Conclusions

- The project demonstrates that tub racing can return as a digital experience that is fun, technically feasible, and meaningful to SPSU alumni engagement.
- Just as importantly, the team now has a stronger multiplayer foundation, clearer documentation needs, and a direction for future teams to build on.

## Future Work

- Finalize bug fixes and smaller enhancements this semester.
- Potential additions: synced kart customization, UI polish, new tracks, and competitive ranking.

## Keys to Success

- Clear communication throughout the project
- Tracked deliverables and deadlines
- Aligned work with sponsor expectations
- Defined roles and accountability
- Used sponsor feedback to refine the build

## Project Information and Acknowledgements

### Team Members

Joshua Young  
Thomas Powell  
William Pitts  
William Urvan

### Course/Stack

4850-02 | Spring 2026  
Unity + Steam + GoNet  
Minimum hardware:  
3GB and 8GB RAM

### Industry Sponsor

Willaim McKenna  
Shaun Sheppard

### Contact Information

